Machine learning

Probabilistic models and learning

Thomas D. Nielsen

Resources

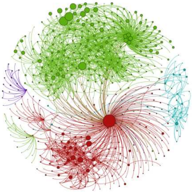
Probabilistic Machine Learning: An introduction by Kevin Murphy:

`https://probml.github.io/pml-book/book1.html`

Bayesian networks and Decision Graphs by Finn V. Jensen and Thomas Dyhre Nielsen

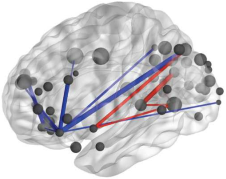
Build, Compute, Critique, Repeat: Data Analysis with Latent Variable Models by David M. Blei

Examples



Communities discovered in a 3.7M node network of U.S. Patents

[Gopalan and Blei, PNAS 2013]

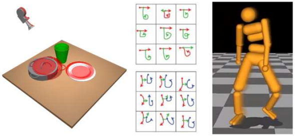


Neuroscience analysis of 220 million fMRI measurements

[Manning et al., PLOS ONE 2014]

Analysis of 1.7M taxi trajectories, in Stan

[Kucukelbir et al., 2016]



Scenes, concepts and control.

[Eslami et al., 2016, Lake et al. 2015]

Images borrowed from David Blei et al.: *Variational Inference: Foundations and Modern Methods* (NeurIPS Tutorial, 2016)

Common ground in the examples:

- Use a **probabilistic model**, typically learned from data (using **statistical learning techniques**)
- Apply probabilistic inference algorithms to use models for prediction (**classification**, **regression**), structure analysis (**clustering**, **segmentation**)

Need: probabilistic models that . . .

- can represent models for high-dimensional state spaces
- support efficient learning and inference techniques

Probabilistic Graphical Models . . .

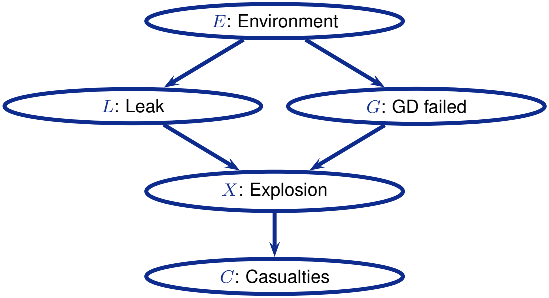
- support a structured specification of high-dimensional distributions in terms of low-dimensional factors
- structured representation can be exploited for efficient learning and inference algorithms (sometimes ...)
- graphical representation gives human-friendly design and description possibilities

Advantages of probabilistic/statistical methods:

- Principled quantification of prediction uncertainties
- Robust and principled techniques to deal with incomplete information, missing data.
- Provides a general model of the domain that can in principle be used to answer any inference query.
- Enables integration of domain knowledge.
- Transparent and explainable.

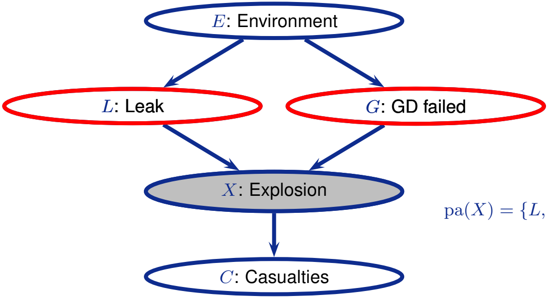
Brief recap from the Machine Intelligence course ...

A simple example: "Explosion"



$p(E, L, G, X, C)$

A simple example: “Explosion”



$\mathrm{pa}(X) = \{L, G\}$

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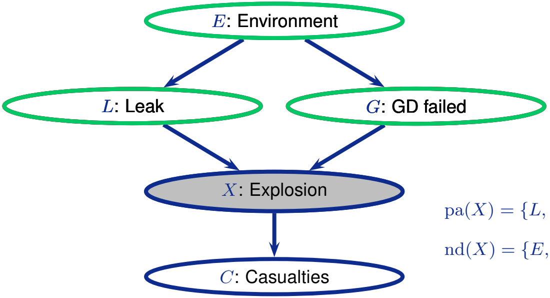

$\mathrm{pa}(X) = \{L, G\}$

$\mathrm{nd}(X) = \{E, L, G\}$

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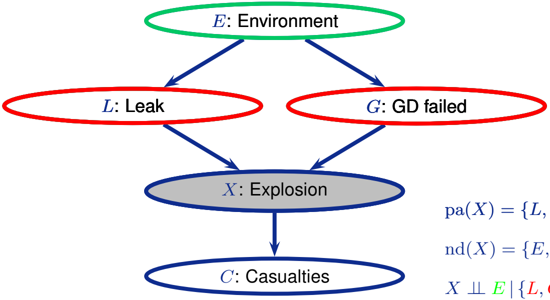

$\mathrm{pa}(X) = \{L, G\}$

$\mathrm{nd}(X) = \{E, L, G\}$

$X \perp\!\!\!\perp E \,|\, \{L, G\}$

Other d-sep. rules: Jensen&Nielsen (07)

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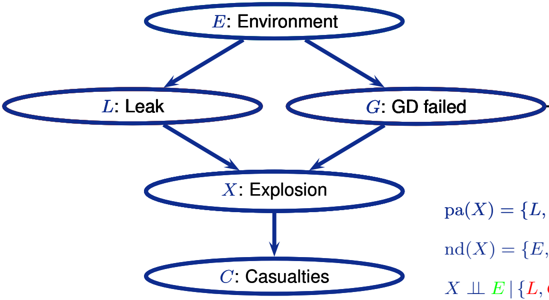


G	$E =$`hostile`	$E =$`normal`
`yes`	$\lambda_H \cdot \tau/2$	$\lambda_N \cdot \tau/2$
`no`	$1 - \lambda_H \cdot \tau/2$	$1 - \lambda_N \cdot \tau/2$

$p(G \,|\, \mathrm{pa}(G))$

$\mathrm{pa}(X) = \{L, G\}$

$\mathrm{nd}(X) = \{E, L, G\}$

$X \perp\!\!\!\perp E \,|\, \{L, G\}$ $\big($Hence, $p(X \,|\, E, L, G) = P(X \,|\, L, G)\big)$

Other d-sep. rules: Jensen&Nielsen (07)

$$p(E, L, G, X, C) = p(E) \cdot p(L \,|\, E) \cdot p(G \,|\, E, L) \cdot p(X \,|\, E, L, G) \cdot p(C \,|\, E, L, G, X)$$

$$= p(E) \cdot p(L \,|\, E) \cdot p(G \,|\, E) \quad \cdot p(X \,|\, L, G) \quad \cdot p(C \,|\, X)$$

Markov properties ⇔ Factorisation property

Bayesian network syntax

Let $X_1, \ldots, X_n$ be a collection of random variables. A Bayesian network over $X_1, \ldots, X_n$ consists of

- a directed acyclic graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ whose nodes $\mathcal{V}$ are the variables $X_1, \ldots, X_n$
- a set of local conditional distributions, $\mathcal{P} = \{p(X_i \mid \mathrm{pa}(X_i)), X_i \in \mathcal{V}\}$, where $\mathrm{pa}(X_i)$ are the parents of X_i in $\mathcal{G}$ as defined by the edges $\mathcal{E}$.

Bayesian network semantics

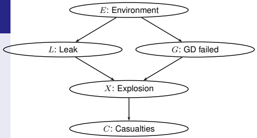
A Bayesian network $\mathcal{N}$ with nodes $X_1, \ldots, X_n$ defines a joint distribution

$$p_{\mathcal{N}}(X_1, \ldots, X_n) = \prod_{i=1}^{n} p(X_i \mid \mathrm{pa}(X_i))$$

Probabilistic inference - Example

Probability propagation

$$\left.\begin{array}{c}\text{Bayesian network}\\ +\\ \text{Evidence: } \mathbf{X}_E = \mathbf{x}_e\end{array}\right\} \Rightarrow p(\mathbf{X}_Q|\mathbf{x}_e)?$$



Example: Calculate $p(X = x \,|\, G = g) = \frac{p(X=x, G=g)}{p(g)}$

$$\begin{aligned} p(x,g) &= \sum_e \sum_l \sum_c p(E=e, L=l, g, X=x, C=c) \\ &= \sum_e \sum_l \sum_c p(e) \cdot p(l \,|\, e) \cdot p(g \,|\, e) \cdot p(x \,|\, l, g) \cdot p(c \,|\, x) \\ &= \sum_e p(e) \cdot p(g \,|\, e) \sum_l p(l \,|\, e) \cdot p(x \,|\, l, g) \sum_c p(c \,|\, x) \end{aligned}$$

Probability propagation

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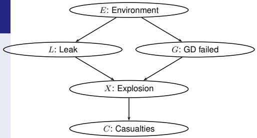

Example: Calculate $p(X = x \,|\, G = g) = \frac{p(X=x, G=g)}{p(g)}$

$$p(x, g) = \sum_e p(e) \cdot p(g \,|\, e) \sum_l p(l \,|\, e) \cdot p(x \,|\, l, g)$$

$$p(X = x \,|\, G = g) = \frac{p(x, g)}{\sum_{x'} p(X = x', g)} \propto p(x, g)$$

Probabilistic inference - Example

Probability propagation

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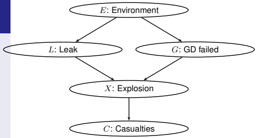

Operations to calculate $P(\mathbf{X}_Q \mid \mathbf{x}_e)$

Restriction: Restrict domain of a potential (e.g., $p(g|E)$ from $p(G|E)$)

Combination: Multiplication of potentials (e.g., $p(l \mid e) \cdot p(X \mid l, g)$)

Marginalisation: Sum/integrate out a variable from a potential, e.g., the operation $\sum_l p(l \mid e) \cdot p(x \mid l, g)$, which removes L from the potential over $\{L, X, E, G\}$ and results in $p(x \mid e, g)$ over $\{X, E, G\}$.

Parameter learning

Thumbtack example

We have tossed a thumb tack 100 times. It has landed pin up 80 times, and we now look for the probability θ of pin up ($\rightsquigarrow$ model that best fits the observations/data):

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We have tossed a thumb tack 100 times. It has landed pin up 80 times, and we now look for the probability θ of pin up ($\rightsquigarrow$ model that best fits the observations/data):

We can measure how well a model fits the data using the likelihood:

$$\begin{aligned} p(\text{data}|\theta) &= p(\text{pin up}, \text{pin up}, \text{pin down}, \ldots, \text{pin up}|\theta) \\ &= p(\text{pin up}|\theta) \cdot p(\text{pin up}|\theta) \cdot p(\text{pin down}|\theta) \cdot \ldots \cdot p(\text{pin up}|\theta) \\ &= \prod_{i=1}^{100} p(x_i \,|\, \theta) \end{aligned}$$

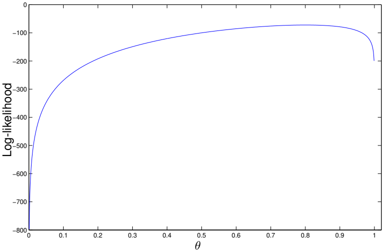
The data is here assumed to have been sampled independently from the same distributions. This is known as the iid assumption (independent and identically distributed).

Thumbtack example

We have tossed a thumb tack 100 times. It has landed pin up 80 times, and we now look for the probability θ of pin up ($\rightsquigarrow$ model that best fits the observations/data):

We select the parameter $\hat{\theta}$ that maximizes:

$$\begin{aligned}\hat{\theta} &= \arg\max_{\theta} p(\text{data}|\theta) \left(\text{or } \log p(\text{data}|\theta)\right)\\ &= \arg\max_{\theta} \prod_{i=1}^{100} p(x_i|\theta)\\ &= \arg\max_{\theta} \theta^{80}(1-\theta)^{20}.\end{aligned}$$



Thumbtack example

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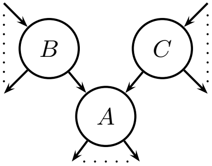
By setting:

$$\frac{d}{d\theta}\theta^{80}(1-\theta)^{20} = 0 \qquad \left(\text{or } \frac{d}{d\theta}\log(\theta^{80}(1-\theta)^{20}) = 0\right)$$

we get the *maximum likelihood estimate*:

$$\hat{\theta} = 0.8.$$

For the Bayesian network



		$B=t$		$B=f$	
		$C=y$	$C=n$	$C=y$	$C=n$
	a_1	θ_{a_1ty}	θ_{a_1tn}	θ_{a_1fy}	θ_{a_1fn}
A	a_2	θ_{a_2ty}	θ_{a_2tn}	θ_{a_2fy}	θ_{a_2fn}
	a_3	θ_{a_3ty}	θ_{a_3tn}	θ_{a_3fy}	θ_{a_3fn}

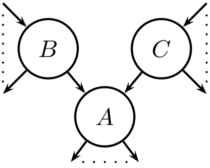
$p(A|B,C)$ (with $\theta_{a_3\cdot\cdot} = 1 - \theta_{a_1\cdot\cdot} + \theta_{a_2\cdot\cdot}$)

and with complete data instances $\mathcal{D} = (\mathbf{x}_1, \ldots, \mathbf{x}_N)$, we look for the parameters $\hat{\boldsymbol{\theta}}$ that maximizes

$$L(\boldsymbol{\theta} \mid \mathcal{D}) = \prod_{i=1}^{N} p(\mathbf{x}_i \mid \boldsymbol{\theta}) \qquad \text{or equivalently} \qquad LL(\boldsymbol{\theta} \mid \mathcal{D}) = \sum_{i=1}^{N} \log p(\mathbf{x}_i \mid \boldsymbol{\theta})$$

Maximum likelihood parameter estimation in BNs

For the Bayesian network



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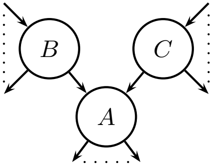
From the chain rule we have that

$$LL(\boldsymbol{\theta} \mid \mathcal{D}) = \sum_{i=1}^{N} \log p(\mathbf{x}_i \mid \boldsymbol{\theta}) = \sum_{i=1}^{N} \sum_{X \in \mathcal{V}} \log p(X \mid \mathrm{pa}(X), \boldsymbol{\theta})(\mathbf{x}_i)$$

hence, the maximization of $LL(\boldsymbol{\theta} \mid \mathcal{D})$ reduces to a set of independent maximizations of

$$\sum_{i=1}^{N} \log p(X \mid \mathrm{pa}(X), \boldsymbol{\theta})(\mathbf{x}_i).$$

For the Bayesian network



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Let

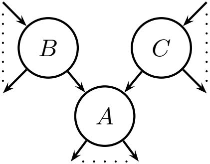
$$N(A=a, B=b, C=c) = |\{\mathbf{x} \in \mathcal{D} | A=a, B=b, C=c \text{ in } \mathbf{x}\}|.$$

To find the maximum likelihood estimate $\hat{\theta}_{a_2tn} = \hat{p}(A=a_2 \,|\, B=t, C=n)$ we calculate:

$$\hat{p}(A=a_2 \,|\, B=t, C=n) =$$

Maximum likelihood parameter estimation in BNs

For the Bayesian network



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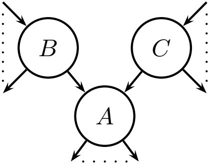

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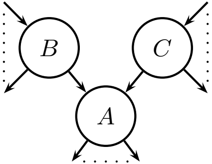
For the Bayesian network



$$\begin{array}{cc|cccc} & & \multicolumn{2}{c}{B=t} & \multicolumn{2}{c}{B=f} \\ & & C=y & C=n & C=y & C=n \\ \hline & a_1 & \theta_{a_1ty} & \theta_{a_1tn} & \theta_{a_1fy} & \theta_{a_1fn} \\ A & a_2 & \theta_{a_2ty} & \theta_{a_2tn} & \theta_{a_2fy} & \theta_{a_2fn} \\ & a_3 & \theta_{a_3ty} & \theta_{a_3tn} & \theta_{a_3fy} & \theta_{a_3fn} \end{array}$$

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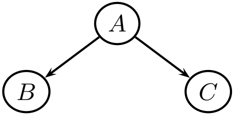
$$= \frac{N(A=a_2, B=t, C=n)}{N(B=t, C=n)}.$$

In general, with *complete data* you get a maximum likelihood estimate as the fraction of counts over the total number of counts.

Maximum likelihood estimation in BNs: Exercise

Model structure:



Data:

	A	B	C
1	*yes*	*pos*	*pos*
2	*yes*	*pos*	*neg*
3	*no*	*neg*	*neg*
4	*no*	*neg*	*pos*
5	*yes*	*neg*	*pos*

Parameter estimates:

A	
yes	
no	

	A	
B	*yes*	*no*
pos		
neg		

	A	
C	*yes*	*no*
pos		
neg		

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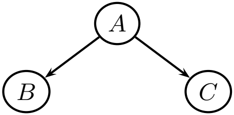

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3	*no*	*neg*	*neg*
4	*no*	*neg*	*pos*
5	*yes*	*neg*	*pos*

Parameter estimates:

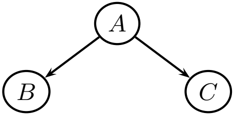
A	
yes	3/5
no	

B	A: *yes*	A: *no*
pos		
neg		

C	A: *yes*	A: *no*
pos		
neg		

Maximum likelihood estimation in BNs: Exercise

Model structure:



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3	*no*	*neg*	*neg*
4	*no*	*neg*	*pos*
5	*yes*	*neg*	*pos*

Parameter estimates:

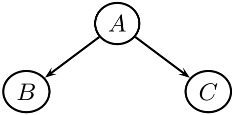
A	
yes	3/5
no	2/5

	A	
B	*yes*	*no*
pos		
neg		

	A	
C	*yes*	*no*
pos		
neg		

Maximum likelihood estimation in BNs: Exercise

Model structure:



Data:

	A	B	C
1	*yes*	*pos*	*pos*
2	*yes*	*pos*	*neg*
3	*no*	*neg*	*neg*
4	*no*	*neg*	*pos*
5	*yes*	*neg*	*pos*

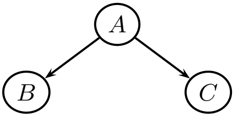
Parameter estimates:

A	
yes	3/5
no	2/5

	A	
B	*yes*	*no*
pos	2/3	
neg		

	A	
C	*yes*	*no*
pos		
neg		

Model structure:



Data:

	A	B	C
1	*yes*	*pos*	*pos*
2	*yes*	*pos*	*neg*
3	*no*	*neg*	*neg*
4	*no*	*neg*	*pos*
5	*yes*	*neg*	*pos*

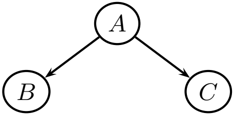
Parameter estimates:

A	
yes	3/5
no	2/5

	A	
B	*yes*	*no*
pos	2/3	
neg	1/3	

	A	
C	*yes*	*no*
pos		
neg		

Model structure:



Data:

	A	B	C
1	*yes*	*pos*	*pos*
2	*yes*	*pos*	*neg*
3	*no*	*neg*	*neg*
4	*no*	*neg*	*pos*
5	*yes*	*neg*	*pos*

Parameter estimates:

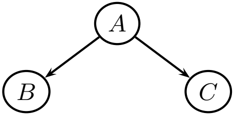
A	
yes	3/5
no	2/5

	A	
B	*yes*	*no*
pos	2/3	0/2
neg	1/3	

	A	
C	*yes*	*no*
pos		
neg		

Maximum likelihood estimation in BNs: Exercise

Model structure:



Data:

	A	B	C
1	*yes*	*pos*	*pos*
2	*yes*	*pos*	*neg*
3	*no*	*neg*	*neg*
4	*no*	*neg*	*pos*
5	*yes*	*neg*	*pos*

Parameter estimates:

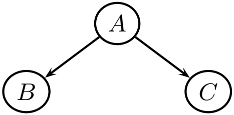
A	
yes	3/5
no	2/5

	A	
B	*yes*	*no*
pos	2/3	0/2
neg	1/3	2/2

	A	
C	*yes*	*no*
pos		
neg		

Maximum likelihood estimation in BNs: Exercise

Model structure:



Data:

	A	B	C
1	*yes*	*pos*	*pos*
2	*yes*	*pos*	*neg*
3	*no*	*neg*	*neg*
4	*no*	*neg*	*pos*
5	*yes*	*neg*	*pos*

Parameter estimates:

A	
yes	3/5
no	2/5

	A	
B	*yes*	*no*
pos	2/3	0/2
neg	1/3	2/2

	A	
C	*yes*	*no*
pos	2/3	1/2
neg	1/3	1/2

MLE for the univariate Gausssian distribution

Assume that we have seen N samples $\{x_1, \ldots, x_N\}$ from a Gaussian distribution $f(x|\mu, \sigma^2)$:

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

MLE for the univariate Gausssian distribution

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$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

From the log likelihood:

$$LL(\mu, \sigma^2) = \frac{-1}{2\sigma^2}\sum_{i=1}^{N}(x_i - \mu)^2 - \frac{N}{2}\log(2\pi\sigma^2)$$

we find the maximum likelihood estimates:

$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$\hat{\sigma}^2 = \frac{1}{N}\sum_{i=!}^{N}(x_i - \hat{\mu})^2$$

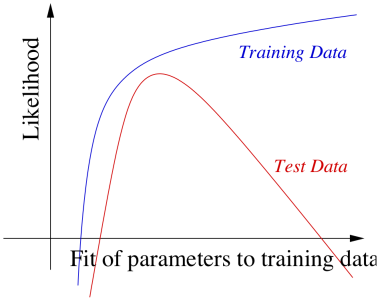
Maximum likelihood parameter in previous example:

$$\hat{p}(B = \textit{pos} \mid A = \textit{no}) = 0$$

If

	A	B	C
…	…	…	…
$\mathbf{x}_6$	*no*	*pos*	*neg*

then $p(\mathbf{x}_6) = 0$, i.e. the model has the worst possible fit for future data:



Parameter Smoothing with Pseudo-Counts

To avoid overfitting, initialize all tables with pseudo-counts (corresponding to virtual data).

E.g. pseudo-count of 1:

	B	
A	*yes*	*no*
pos	1	1
neg	1	1

Now add the actual counts from the data:

	B	
A	*yes*	*no*
pos	1+2	1+0
neg	1+1	1+2

and normalize:

	B	
A	*yes*	*no*
pos	3/5	1/4
neg	2/5	3/4

The value of the pseudo count may also be found by optimization on a separate validation set.

Incomplete data

Missing Values

So far, we always assumed complete data:

Instance	*Outlook*	*Temperature*	*Humidity*	*Wind*	*PlayTennis*
1	*sunny*	*hot*	*high*	*weak*	*no*
2	*sunny*	*hot*	*high*	*strong*	*no*
3	*overcast*	*hot*	*high*	*weak*	*yes*
4	*rain*	*mild*	*high*	*weak*	*yes*
...	...	...	...	...	...

In reality, data sets have missing values:

Instance	*Outlook*	*Temperature*	*Humidity*	*Wind*	*PlayTennis*
1	?	*hot*	*high*	*weak*	*no*
2	*sunny*	*hot*	*high*	*strong*	*no*
3	*overcast*	?	?	*weak*	*yes*
4	*rain*	*mild*	*high*	*weak*	*yes*
...	...	...	...	...	...

When we talk about incomplete data, we usually mean data with missing values.

Ad hoc solutions (e.g., delete instances with missing values, impute values) come with their own problems ...

How is the data missing?

We need to take into account <u>how</u> the data is missing:

Missing completely at random The probability that a value is missing is independent of both the observed and unobserved values.

Missing at random The probability that a value is missing depends only on the observed values (missing at random).

Non-ignorable Neither MAR nor MCAR.

What is the type of missingness in the following scenarios?

- In an exit poll, where an extreme right-wing party is running for parlament.
- In a database containing the results of two tests, where the second test has only performed (as a "backup test") when the result of the first test was negative.
- In a monitoring system that is not completely stable and where some sensor values are not stored properly.

Marginal Likelihood

Marginal likelihood: identify probability of incomplete sample Y_i with the marginal probability of observed variables.

Example

$N = 1$, $s_1 = (38, normal, ?, neg, ?)$. Then

$$\begin{aligned} L(\theta) &= P_\theta(V_1 = 38, V_2 = normal, V_4 = neg) \\ &= \sum_{\substack{v_3 \in \{pos, neg\} \\ v_5 \in \{yes, no\}}} P_\theta(V_1 = 38, V_2 = normal, V_3 = v_3, V_4 = neg, V_5 = v_5) \end{aligned}$$

Maximizing the marginal likelihood

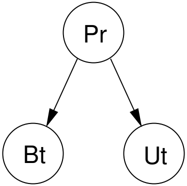
- If the data are *misssing at random* then we can use the marginal likelihood similar to the way we use the standard likelihood for complete data.
- In general a parameter θ maximizing the marginal likelihood cannot be computed in closed form. Need iterative optimization techniques.

The EM algorithm

Idea:

- Use hypothetical completions of data sets obtained from current estimate for θ.
- Apply maximum likelihood inference to completed data to obtain new estimate for θ.

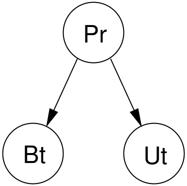
The EM algorithm: An Example



Cases	Pr	Bt	Ut
1.	?	pos	pos
2.	yes	neg	pos
3.	yes	pos	?
4.	yes	pos	neg
5.	?	neg	?

Estimate the required probability distributions for the network

The EM algorithm: An Example



Cases	Pr	Bt	Ut
1.	?	pos	pos
2.	yes	neg	pos
3.	yes	pos	?
4.	yes	pos	neg
5.	?	neg	?

Estimate the required probability distributions for the network

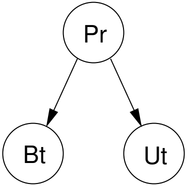
If the database was complete we would estimate the required probabilities, $P(Pr)$, $P(Ut \mid Pr)$ and $P(Bt \mid Pr)$ as:

$$P(Pr = yes) = \frac{N(Pr = yes)}{N} \qquad P(Ut = yes \mid Pr = yes) = \frac{N(Ut = yes, Pr = yes)}{N(Pr = yes)}$$

$$P(Ut = yes \mid Pr = yes) = \frac{N(Ut = yes, Pr = yes)}{N(Pr = yes)}$$

So estimating the probabilities would basically be a counting problem!

The EM algorithm: An Example



Cases	Pr	Bt	Ut
1.	?	pos	pos
2.	yes	neg	pos
3.	yes	pos	?
4.	yes	pos	neg
5.	?	neg	?

Estimate the required probability distributions for the network

To estimate $P(Pr)$ from the database above:

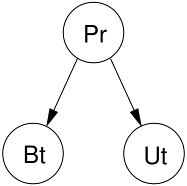
- Case 2, 3 and 4 contributes with a value 1
- Case 1 contributes with $P(Pr = yes|Bt = pos, Ut = pos)$.
- Case 5 contributes with $P(Pr = yes|Bt = neg)$.

To find these probabilities we assume some initial distributions, $P_0(\cdot)$.

$\rightsquigarrow$ We are basically calculating the expectation for $N(Pr = yes)$:

$$\mathbb{E}[N(Pr = yes)] = \sum_{i=1}^{5} P_0(Pr = yes|\mathbf{d}_i)$$

The EM algorithm: An Example



Cases	Pr	Bt	Ut
1.	?	pos	pos
2.	yes	neg	pos
3.	yes	pos	?
4.	yes	pos	neg
5.	?	neg	?

Estimate the required probability distributions for the network

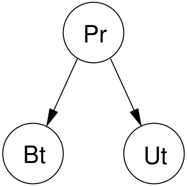
Using $P_0(Pr) = (0.5, 0.5)$, $P_0(Bt \mid Pr = yes) = (0.5, 0.5)$ etc., as starting distributions:

$$\begin{aligned}
\mathbb{E}[N(Pr = yes)] &= P_0(Pr = yes \mid Bt = Ut = pos) + 1 + 1 + 1 + P_0(Pr = yes \mid Bt = neg) \\
&= 0.5 + 1 + 1 + 1 + 0.5 = 4 \\
\mathbb{E}[N(Pr = no)] &= P_0(Pr = no \mid Bt = Ut = pos) + 0 + 0 + 0 + P_0(Pr = no \mid Bt = neg) \\
&= 0.5 + 0 + 0 + 0 + 0.5 = 1
\end{aligned}$$

So we e.g. get:

$$\hat{P}_1(Pr = yes) = \frac{\mathbb{E}[N(Pr = yes)]}{N} = \frac{4}{5} = 0.8$$

The EM algorithm: An Example



Cases	Pr	Bt	Ut
1.	?	pos	pos
2.	yes	neg	pos
3.	yes	pos	?
4.	yes	pos	neg
5.	?	neg	?

Estimate the required probability distributions for the network

To estimate $\hat{P}_1(Ut \mid Pr) = \mathbb{E}[N(Ut, Pr)] / \mathbb{E}[N(Pr)]$ we need:

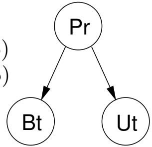
$$\begin{aligned}
\mathbb{E}[N(Ut = p, Pr = y)] =& P_0(Ut = p, Pr = y \mid Bt = Ut = p) + 1 \\
&+ P_0(Ut = p, Pr = y \mid Bt = p, Pr = y) + 0 + P_0(Ut = p, Pr = y \mid Bt = n) \\
=& 0.5 + 1 + 0.5 + 0 + 0.25 = 2.25 \\
\mathbb{E}[N(Pr = yes)] =& P_0(Pr = yes \mid Bt = Ut = pos) + 1 + 1 + 1 + P_0(Pr = yes \mid Bt = neg) \\
=& 0.5 + 1 + 1 + 1 + 0.5 = 4
\end{aligned}$$

So we get:

$$\hat{P}_1(Ut = pos \mid Pr = yes) = \frac{\mathbb{E}[N(Ut = p, Pr = y)]}{\mathbb{E}[N(Pr = yes)]} = \frac{2.25}{4} = 0.5625$$

Learning parameters with incomplete data

$P_0(Pr) = (0.5, 0.5)$
$P_0(Ut = pos \mid Pr) = (0.5, 0.5)$
$P_0(Bt = pos \mid Pr) = (0.5, 0.5)$

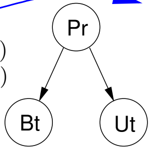


Cases	Pr	Bt	Ut
1.	?	pos	pos
2.	yes	neg	pos
3.	yes	pos	?
4.	yes	pos	neg
5.	?	neg	?

Learning parameters with incomplete data

E-step 1

$P_0(Pr) = (0.5, 0.5)$
$P_0(Ut = pos \mid Pr) = (0.5, 0.5)$
$P_0(Bt = pos \mid Pr) = (0.5, 0.5)$



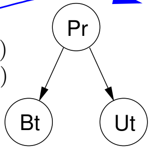
$$\mathbb{E}_0[N(Pr)] = (4, 1)$$
$$\mathbb{E}_0[N(Ut = pos, Pr)] = (2.25, \; 0.5 + 0 + 0 + 0 + 0.25)$$
$$\mathbb{E}_0[N(Bt = pos, Pr)] = (0.5 + 0 + 1 + 1 + 0 = 2.5 \; , 0.5 + 0 + 0 + 0 + 0 = 0.5)$$

Cases	Pr	Bt	Ut
1.	?	pos	pos
2.	yes	neg	pos
3.	yes	pos	?
4.	yes	pos	neg
5.	?	neg	?

Learning parameters with incomplete data

E-step 1

$P_0(Pr) = (0.5, 0.5)$
$P_0(Ut = pos \mid Pr) = (0.5, 0.5)$
$P_0(Bt = pos \mid Pr) = (0.5, 0.5)$



$\mathbb{E}_0[N(Pr)] = (4, 1)$
$\mathbb{E}_0[N(Ut = pos, Pr)] = (2.25, 0.5 + 0 + 0 + 0 + 0.25)$
$\mathbb{E}_0[N(Bt = pos, Pr)] = (0.5 + 0 + 1 + 1 + 0 = 2.5, 0.5 + 0 + 0 + 0 + 0 = 0.5)$

M-step 2

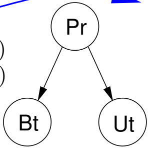
$P_1(Pr) = (\frac{4}{5}, \frac{1}{5})$
$P_1(Ut = pos \mid Pr) = (\frac{2.25}{4}, \frac{0.75}{1})$
$P_1(Bt = pos \mid Pr) = (\frac{2.5}{4}, \frac{0.5}{1})$

Cases	Pr	Bt	Ut
1.	?	pos	pos
2.	yes	neg	pos
3.	yes	pos	?
4.	yes	pos	neg
5.	?	neg	?

Learning parameters with incomplete data

E-step 1

$P_0(Pr) = (0.5, 0.5)$
$P_0(Ut = pos \mid Pr) = (0.5, 0.5)$
$P_0(Bt = pos \mid Pr) = (0.5, 0.5)$

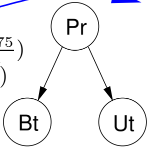


$\mathbb{E}_0[N(Pr)] = (4, 1)$
$\mathbb{E}_0[N(Ut = pos, Pr)] = (2.25, 0.5 + 0 + 0 + 0 + 0.25)$
$\mathbb{E}_0[N(Bt = pos, Pr)] = (0.5 + 0 + 1 + 1 + 0 = 2.5, 0.5 + 0 + 0 + 0 + 0 = 0.5)$

M-step 2

E-step 3

$P_1(Pr) = (\frac{4}{5}, \frac{1}{5})$
$P_1(Ut = pos \mid Pr) = (\frac{2.25}{4}, \frac{0.75}{1})$
$P_1(Bt = pos \mid Pr) = (\frac{2.5}{4}, \frac{0.5}{1})$



$\mathbb{E}_1[N(Pr)] = (\quad , \quad)$
$\mathbb{E}_1[N(Ut = pos, Pr)] = (\quad , \quad)$
$\mathbb{E}_1[N(Bt = pos, Pr)] = (\quad , \quad)$

Cases	Pr	Bt	Ut
1.	?	pos	pos
2.	yes	neg	pos
3.	yes	pos	?
4.	yes	pos	neg
5.	?	neg	?

Learning parameters with incomplete data

E-step 1

$P_0(Pr) = (0.5, 0.5)$
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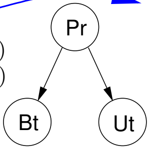

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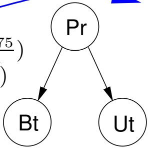
$\mathbb{E}_0[N(Ut = pos, Pr)] = (2.25,$
$0.5 + 0 + 0 + 0 + 0.25)$
$\mathbb{E}_0[N(Bt = pos, Pr)] = (0.5 + 0 + 1 + 1 + 0 = 2.5$
$, 0.5 + 0 + 0 + 0 + 0 = 0.5)$

M-step 2

E-step 3

$P_1(Pr) = (\frac{4}{5}, \frac{1}{5})$
$P_1(Ut = pos \mid Pr) = (\frac{2.25}{4}, \frac{0.75}{1})$
$P_1(Bt = pos \mid Pr) = (\frac{2.5}{4}, \frac{0.5}{1})$



$\mathbb{E}_1[N(Pr)] = (\quad , \quad)$
$\mathbb{E}_1[N(Ut = pos, Pr)] = (\quad , \quad)$
$\mathbb{E}_1[N(Bt = pos, Pr)] = (\quad , \quad)$

M-step 4

$P_2(Pr) = (\cdot, \cdot)$
$P_2(Ut = pos \mid Pr) = (\cdot, \cdot)$
$P_2(Bt = pos \mid Pr) = (\cdot, \cdot)$

$\vdots$

Until convergence

Cases	Pr	Bt	Ut
1.	?	pos	pos
2.	yes	neg	pos
3.	yes	pos	?
4.	yes	pos	neg
5.	?	neg	?

The EM algorithm: in general

1. Let $\boldsymbol{\theta}^0 = \{\theta_{ijk}\}$ be some start estimates ($P(X_i = j \mid \mathrm{pa}(X_i = k) = \theta_{ijk}$).
2. Repeat until convergence:
   - E-step: For each variable X_i calculate the table of expected counts:

$$\mathbb{E}[N(X_i, \mathrm{pa}(X_i) \mid \mathcal{D}] = \sum_{\mathbf{d} \in \mathcal{D}} P(X_i, \mathrm{pa}(X_i) \mid \mathbf{d}, \boldsymbol{\theta}^t).$$

   - M-step: Use the expected counts as if they were actual counts:

$$\hat{\theta}_{ijk}^{t+1} = \frac{\mathbb{E}[N(X_i = k, \mathrm{pa}(X_i) = j \mid \mathcal{D}]}{\sum_{k=1}^{|\mathrm{sp}(X_i)|} \mathbb{E}[N(X_i = k, \mathrm{pa}(X_i) = j \mid \mathcal{D}]}.$$

Correctness

The sequence of estimates θ generated by the EM algorithm converges to a local maximum (in rare cases: a saddle point) of the marginal likelihood for θ given $\mathbf{s}$.

To avoid sub-optimal local maxima: run EM several times with different starting points.

Bayesian parameter learning

Basis

- Parameters are modeled as random variables and information about them can be included prior to observing data
- By Bayes' rule, the prior information is combined with the likelihood, yielding a posterior distribution that is used for making inferences about the parameter

More formally Parameter learning amounts to calculating the posterior distribution over the parameters given the data $\mathcal{D} = (\mathbf{x}_1, \ldots, \mathbf{x}_N)$:

$$p(\boldsymbol{\theta} \,|\, \mathcal{D}) = \frac{p(\boldsymbol{\theta}, \mathcal{D})}{p(\mathcal{D})} = \frac{p(\boldsymbol{\theta}) p(\mathcal{D} \,|\, \boldsymbol{\theta})}{p(\mathcal{D})}$$

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Basis

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$$p(\mathcal{D} \,|\, \boldsymbol{\theta}) = \prod_{i=1}^{N} p(\mathbf{x}_i \,|\, \boldsymbol{\theta}) \qquad \text{[Assuming independently sampled data]}$$

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$$p(\boldsymbol{\theta} \,|\, \mathcal{D}) = \frac{p(\boldsymbol{\theta}, \mathcal{D})}{p(\mathcal{D})} = \frac{p(\boldsymbol{\theta})p(\mathcal{D} \,|\, \boldsymbol{\theta})}{p(\mathcal{D})}$$

$$p(\mathcal{D} \,|\, \boldsymbol{\theta}) = \prod_{i=1}^{N} p(\mathbf{x}_i \,|\, \boldsymbol{\theta}) \qquad \text{[Assuming independently sampled data]}$$

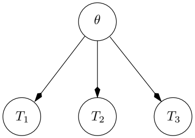
Posterior predictive distribution

For making predictions about future cases we use

$$p(\mathbf{X} \,|\, \mathcal{D}) = \int_{\boldsymbol{\theta}} p(\mathbf{X} \,|\, \boldsymbol{\theta}) p(\boldsymbol{\theta} \,|\, \mathcal{D}) d\boldsymbol{\theta}$$

Complete data: Bayesian estimation

The Bayesian approach:



Using some density function $p(\theta)$ and with $p(\text{pin up} \mid \theta)$ as θ, we look for:

$$p(\theta \mid \text{pin up}) = \frac{p(\text{pin up} \mid \theta)p(\theta)}{p(\text{pin up})}$$

$$= \frac{p(\text{pin up} \mid \theta)}{p(\text{pin up})} = \frac{\theta}{p(\text{pin up})} \qquad [p(\theta) = 1]$$

$$= 2\theta \qquad [p(\text{pin up}) = \int_0^1 \theta d\theta = 1/2]$$

A single best estimate could then be the mean value:

$$\int_0^1 \theta(2\theta)d\theta = 2/3$$

Beta priors

Model and data

Consider the variables $X_1, X_2, \ldots, X_N$ with $P(X_i|\theta) = \theta$:

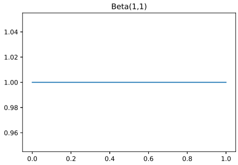
- $X_i = 1$ represents pin up and $X_i = 0$ represents pin down
- $\theta \in [0, 1]$ is the probability of pin up

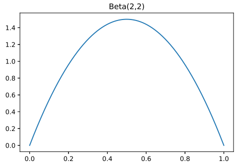
Prior

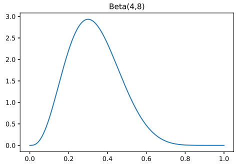
The beta distribution Beta$(\theta \mid a, b)$ over θ:

$$p(\theta) \propto \theta^{a-1}(1 - \theta^{b-1}),$$

where a and b are hyper parameters.







Beta posterior

Model and data

Suppose that $X_1, X_2, \ldots, X_N \sim \text{Ber}(\theta)$

- N_1 are the number of pin up
- N_0 are the number of pin down $(N = N_0 + N_1)$

Beta posterior

Model and data

Suppose that $X_1, X_2, \ldots, X_N \sim \text{Ber}(\theta)$

- N_1 are the number of pin up
- N_0 are the number of pin down $(N = N_0 + N_1)$

Prior

The beta distribution $\text{Beta}(\theta \mid a, b)$ over θ:

$$p(\theta) \propto \theta^{a-1}(1-\theta^{b-1})$$

Beta posterior

Model and data

Suppose that $X_1, X_2, \ldots, X_N \sim \text{Ber}(\theta)$

- N_1 are the number of pin up
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$$p(\theta) \propto \theta^{a-1}(1 - \theta^{b-1})$$

Posterior

$$p(\theta \mid \mathcal{D}) = \text{Beta}(\theta \mid a + N_1, b + N_0)$$

Beta posterior

Model and data

Suppose that $X_1, X_2, \ldots, X_N \sim \text{Ber}(\theta)$

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Posterior

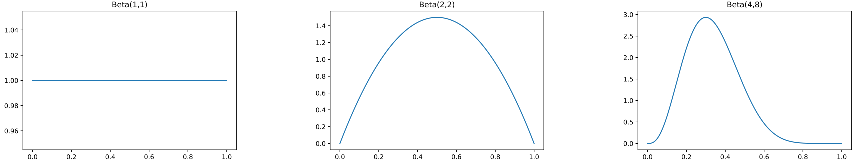
$$p(\theta \mid \mathcal{D}) = \text{Beta}(\theta \mid a + N_1, b + N_0)$$

Posterior predictive distribution

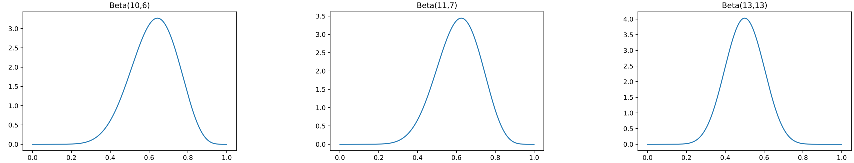
$$\begin{aligned} p(X_{N+1} = 1 \mid \mathcal{D}) &= \int_0^1 p(X_{N+1} = 1 \mid \theta) p(\theta \mid \mathcal{D}) d\theta \\ &= \int_0^1 \theta \text{Beta}(\theta \mid a + N_1, b + N_0) d\theta = \frac{a + N_1}{a + b + N} \end{aligned}$$

Assume that $N_0 = 5$ and $N_1 = 9$.

Prior



Posterior



Learning probabilistic models from data: Bayesian approach

- The method above is also known as *fully Bayesian* approach
  - Parameters are treated as latent variables
  - Parameter learning reduces to inference in graphical models
- Rather than computing the full posterior, we may settle for a point estimate such as

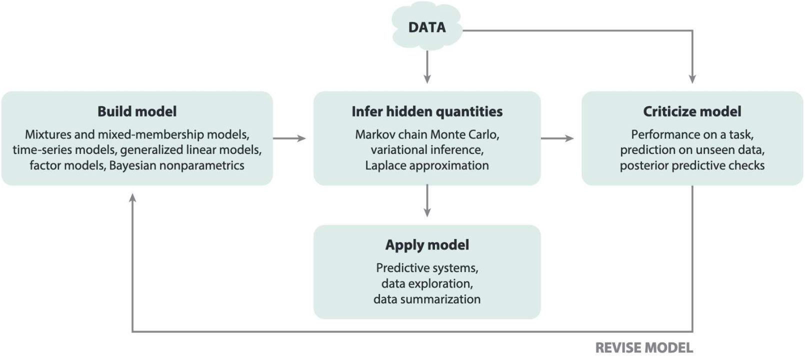
$$\hat{\theta} = \arg\max_{\theta} \log p(x_1, \ldots, x_n|\theta) + \log p(\theta)$$

  known as the (maximum a posteriori) MAP estimator
- Note that we could also choose

$$\hat{\theta} = \arg\max_{\theta} \log p(x_1, \ldots, x_n|\theta)$$

  which is actually the maximum likelihood estimator.

Box's loop



Build, Compute, Critique, Repeat: Data Analysis with Latent Variable Models by David M. Blei